

Fusion-Compatible Budgeted Compact Export with Degradation-Supervised Reliability

for Multimodal Remote Sensing Classification

Core question

Can a multimodal RS model be physically compact and robust to missing/degraded modalities?

Main evidence

Actual MAC budget agreement + robustness under controlled modality states.

Report scope

Submission-oriented story; thesis extensions remain future work.

Research motivation: why lightweight multimodal RS matters

2+

sensor branches

80%

target MAC setting

3

dataset roles

Deployment pressure

Edge/onboard inference needs lower MACs, parameters, memory traffic, and stable behavior.

HSI provides rich spectral signatures but is high-dimensional.

LiDAR or auxiliary sensors add geometry and spatial cues.

Multi-branch fusion improves accuracy but raises cost and failure modes.

A deployable method must report measured compact resources, not only training-time gates.

Problem: resource budget + missing/degraded modality

Budget constraint

The final network should satisfy a target resource ratio ρ measured by actual MACs/params.

Missing modality

A modality can be physically unavailable and must be excluded from fusion competition.

Degraded modality

A modality can remain available but become noisy, low-resolution, or occluded.

Desired behavior

Keep complete-modality accuracy competitive.

Export a physically smaller compact model.

Handle full, main-only, auxiliary-only, and corrupted auxiliary inputs.

Treat reliability as a degradation-supervised diagnostic signal.

Key gap in existing work

Method family	Budget	Compact export	Missing	Degraded	Fusion interface
Structured pruning	Yes	Yes	No	No	Usually single-backbone
Elastic networks	Yes	Partial	No	No	Width-adaptive only
Multimodal RS fusion	No	No	Limited	Limited	Accuracy-first
Missing-modality RS	Limited	No	Yes	Partial	Often heavier
BRM-Net	Yes	Yes	Yes	Yes	Explicitly preserved

Gap 1

Compression rarely respects multimodal fusion interfaces.

Gap 2

Missing-modality methods often rely on reconstruction or heavier modules.

Gap 3

Degraded-but-available modalities are under-tested.

BRM-Net overview



Main design principle: compress internal channels while preserving the terminal fusion feature interface.

Problem formulation

Constrained objective

Deployment target

$\min E[L_{\text{cls}}(f_{\text{theta}}(g(x), a), y)]$
subject to $R(f_{\text{theta}}^c) \leq \rho R(f_{\text{theta}})$

$x^{(1)}, x^{(2)}$: heterogeneous aligned observations such as HSI-LiDAR.

a_m in $\{0,1\}$: availability mask; missing modalities are excluded from fusion.

g in G : controlled degradation: noise, downsampling, occlusion.

f_{theta}^c : physically compact model after structural projection.

Narrow claim

BRM-Net does not claim physical sensor calibration. It evaluates degradation-supervised reliability as a diagnostic fusion signal under controlled modality states.

Budget-aware channel gating

Gate definition

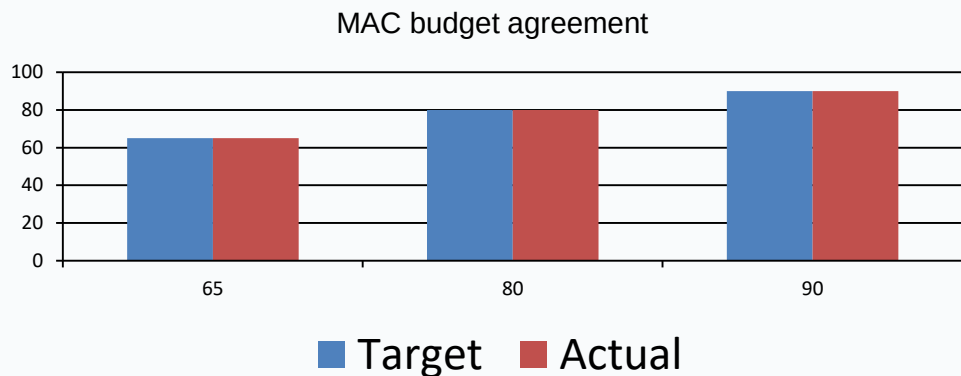
Each gated layer has z_l in $[0,1]^C$. Features are multiplied channel-wise before export.

Training estimate

Hard-concrete probabilities estimate expected channel usage and resource cost.

Export rule

Threshold gates are binarized to meet target MACs while respecting minimum widths.



Resource loss penalizes mismatch between expected and target MAC ratio.

Gating is only an intermediate training mechanism.

The final report evaluates exported compact models.

Fusion-compatible compact export

Why compatibility matters

Independent branch pruning can break the feature dimensions expected by the fusion head.

Structural projection

Remove output channels, dependent input slices, BN parameters, and classifier weights.

Terminal constraint

Preserve fusion-consumed terminal representations through tied or protected gates.

State	Source OA	Compact OA	Delta	MACs
Full	86.15	86.90	+0.75	80.01
HSI only	78.18	80.04	+1.86	80.01
Noise-high	84.26	84.55	+0.29	80.01
Occlusion-50	84.59	85.57	+0.98	80.01

Source gated model and physically exported compact model remain close, supporting the compact-export claim.

Availability-aware reliability fusion

Reliability scores

Each modality produces q_m , a learned quality score used in fusion weighting.

Availability mask

Unavailable modalities receive a large negative logit before softmax.

Fusion weights

$\alpha_m = \text{softmax}(\log q_m + \text{mask}(a_m))$ over usable modalities.

State	BRM-Net	w/o mask	Delta
Full	86.90	87.02	+0.12
HSI only	80.04	76.72	-3.32
LiDAR only	38.04	37.59	-0.45
Occlusion-50	85.57	85.67	+0.10

Interpretation: the mask mainly protects physically missing modalities, not every degraded-but-available case.

Degradation-supervised quality learning

Clean / available

target $t_m = 1$

Missing

target $t_m = 0$

Noise / occlusion

intermediate target

Downsample x4

lower target

Training loss

Objective

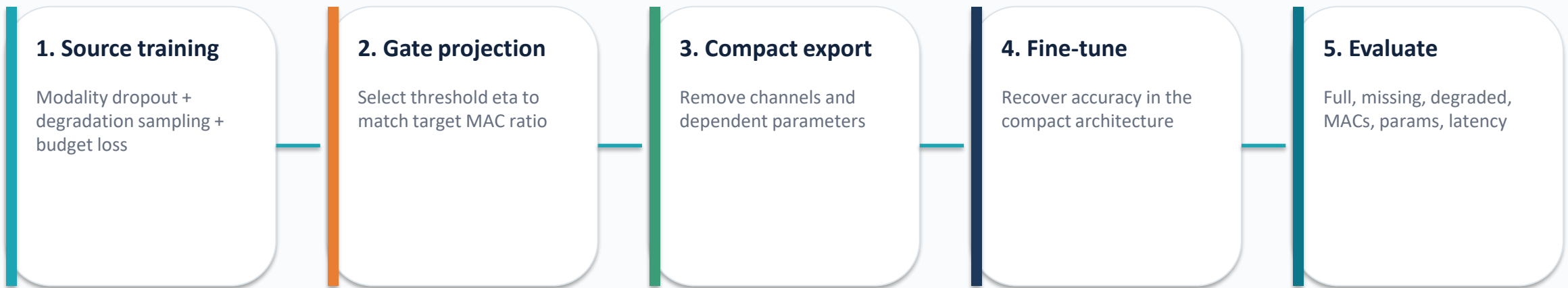
$$L = L_{cls} + \lambda_b L_{bud} + \lambda_q L_{qual}$$

Degraded modalities remain available during fusion.

The reliability branch must learn to down-weight corrupted inputs.

This is not a calibrated physical sensor-quality estimator.

Training and export pipeline



Algorithmic emphasis

The export stage is essential: the deployment model has physically fewer channels rather than merely soft gates in the source network.

Experimental protocol and datasets

Dataset	Role	Train/Test	Why included
Houston2013	Main benchmark	2,832 / 12,197	Official split; full ablation protocol
Trento	External validation	819 / 29,395	Near-saturated HSI-LiDAR accuracy test
MUUFL	Stress test	1,550 / 52,137	Imbalanced scene; weak auxiliary modality

Metrics

OA, AA, Kappa, MAC ratio, parameter ratio, latency where available.

States

Full, main-only, auxiliary-only, noise, downsampling, occlusion.

Reporting rule

Use fixed seeds and avoid leaderboard-style cross-dataset overclaiming.

Budget agreement results

64.96%

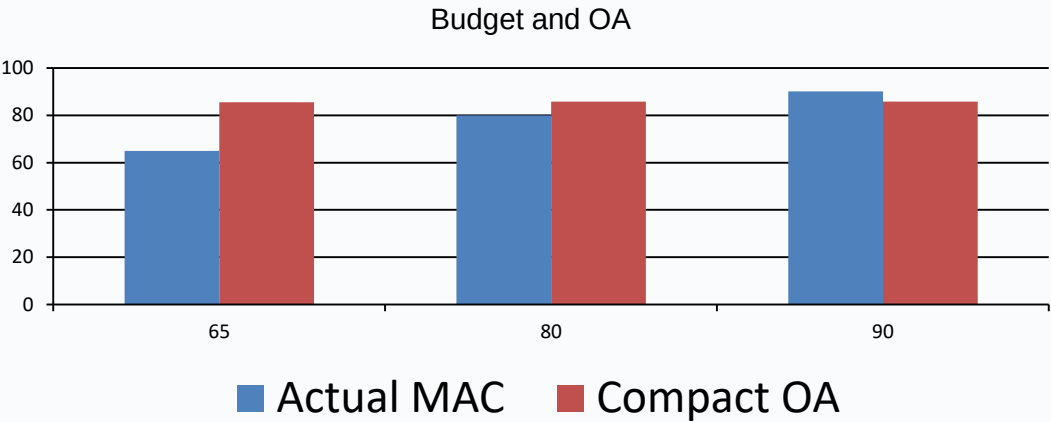
actual MAC at 65 target

80.01%

actual MAC at 80 target

90.07%

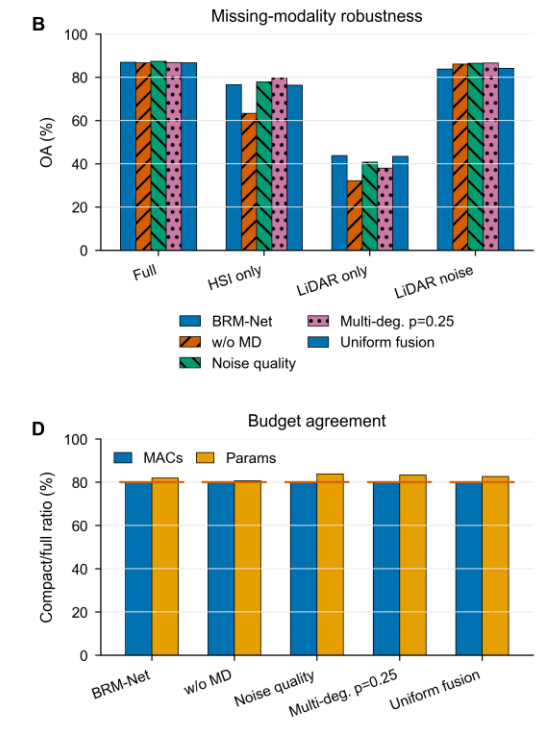
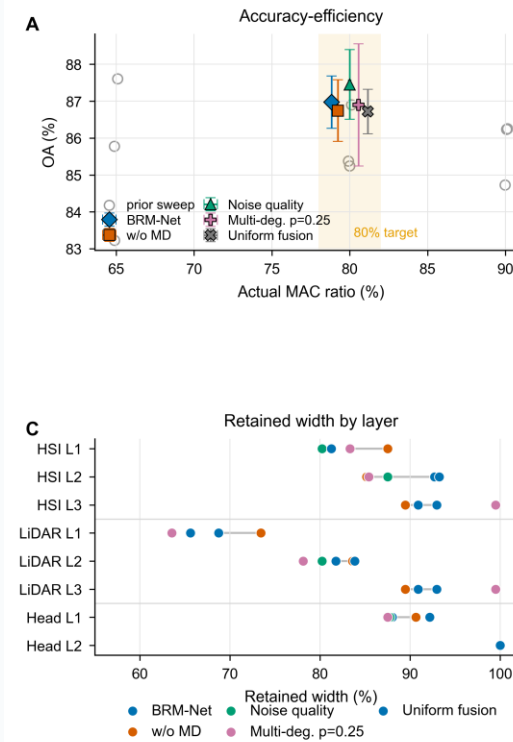
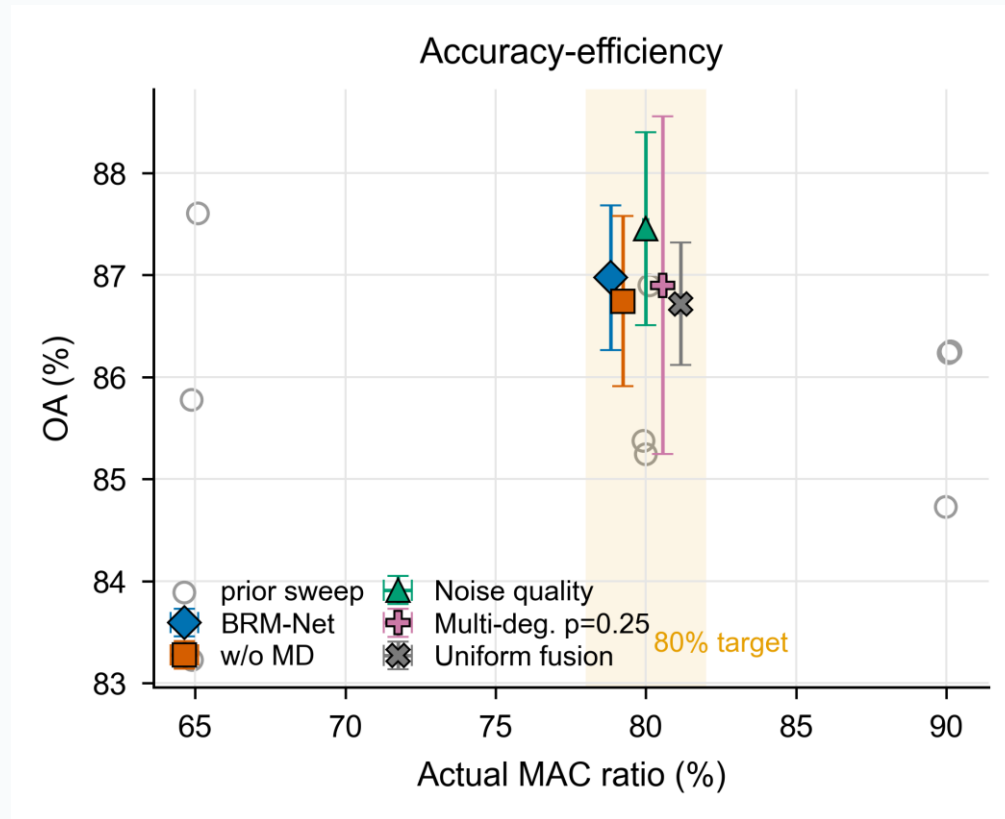
actual MAC at 90 target



Target	Source OA	Compact OA	Params	MACs
65	85.98	85.53	64.79	64.96
80	85.79	85.84	79.92	80.01
90	85.33	85.74	89.78	90.07

Main claim: target budgets can be materialized as measured compact networks.

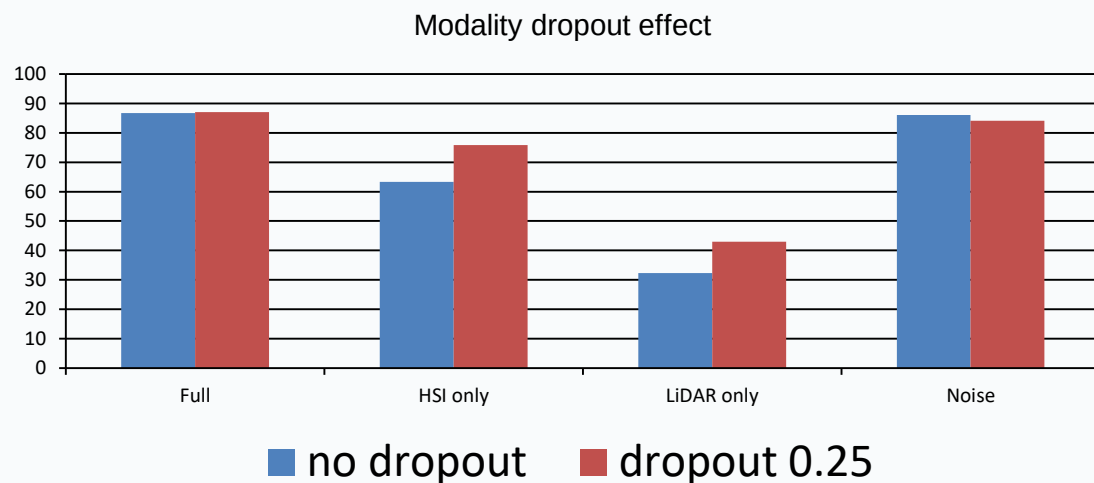
Accuracy-efficiency trade-off



Evidence combines Pareto-style accuracy-efficiency view and the paper overview figure.

Missing-modality robustness

Training	Full	HSI only	LiDAR only	LiDAR noise
no dropout	86.74	63.34	32.27	86.13
dropout 0.25	87.01	75.84	42.95	84.09



Observation

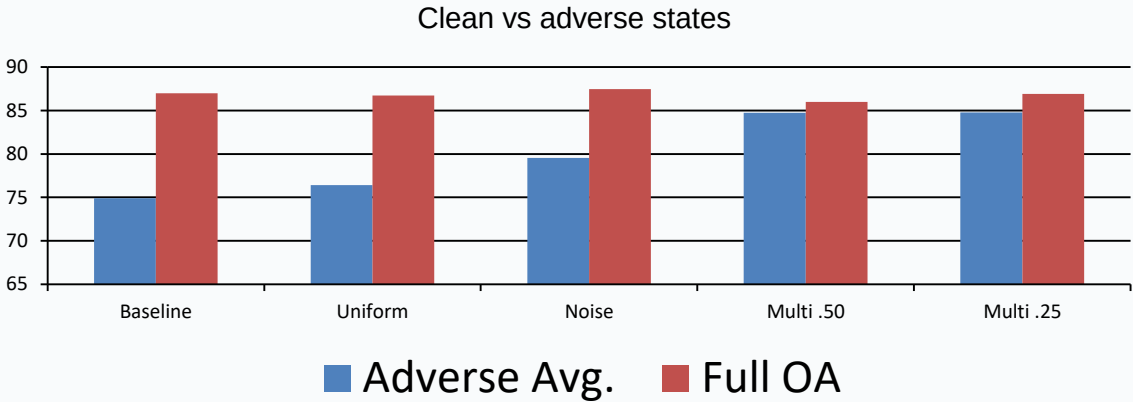
Training-time modality dropout greatly improves single-modality fallback.

Boundary

Availability masking protects missing modalities; degradation needs quality supervision.

Degraded-modality robustness

Method	Full	Noise-high	Down-4	Occ-50	Adverse
Full baseline	86.97	72.08	78.37	74.21	74.89
Uniform fusion	86.72	77.01	78.35	73.87	76.41
Noise quality	87.45	85.15	80.14	73.32	79.54
Multi-deg p=.50	85.99	85.11	84.38	84.66	84.72
Multi-deg p=.25	86.90	84.55	84.15	85.57	84.76

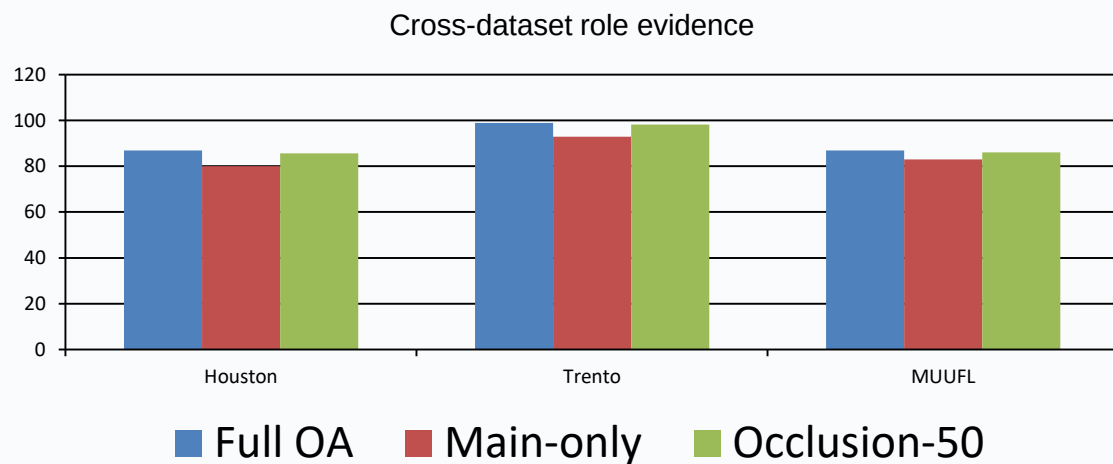


Key result

Adverse-state average improves from 74.89 to 84.76 under the 80% MAC setting.

Multi-dataset evidence: Houston / Trento / MUUFL

Dataset	Full OA	Main-only	Occlusion-50	MACs
Houston2013	86.90 $\pm$ 1.65	80.04 $\pm$ 1.93	85.57 $\pm$ 1.64	80.01
Trento	98.94 $\pm$ 0.94	92.89 $\pm$ 3.15	98.16 $\pm$ 1.87	79.90
MUUFL	86.87 $\pm$ 1.79	83.03 $\pm$ 2.14	85.99 $\pm$ 2.08	79.97



Interpretation

Houston supports the main ablation story; Trento supports efficiency preservation; MUUFL exposes clean-robustness trade-off.

Ablation study

Ablation	What is removed/changed	Observed effect
Uniform fusion	No reliability weighting	Limited corruption robustness
Noise quality	Only noise-supervised quality	Noise improves; occlusion remains weak
Multi-deg p=.50	Stronger degradation schedule	Robust but slightly hurts full OA
Multi-deg p=.25	Light multi-degradation schedule	Best adverse-state average
w/o mask	Disable availability mask	HSI-only OA drops by 3.32 pp

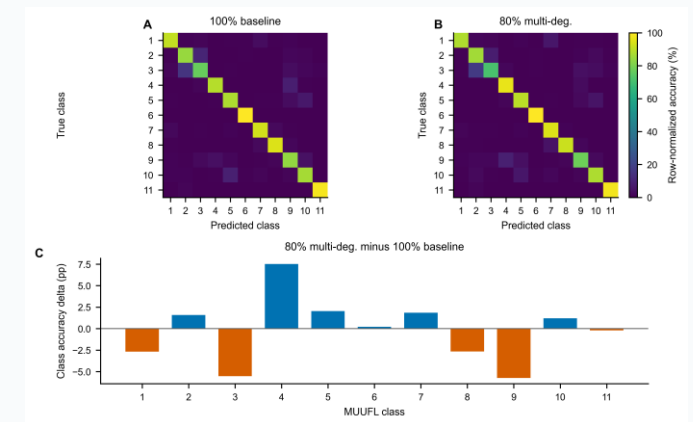
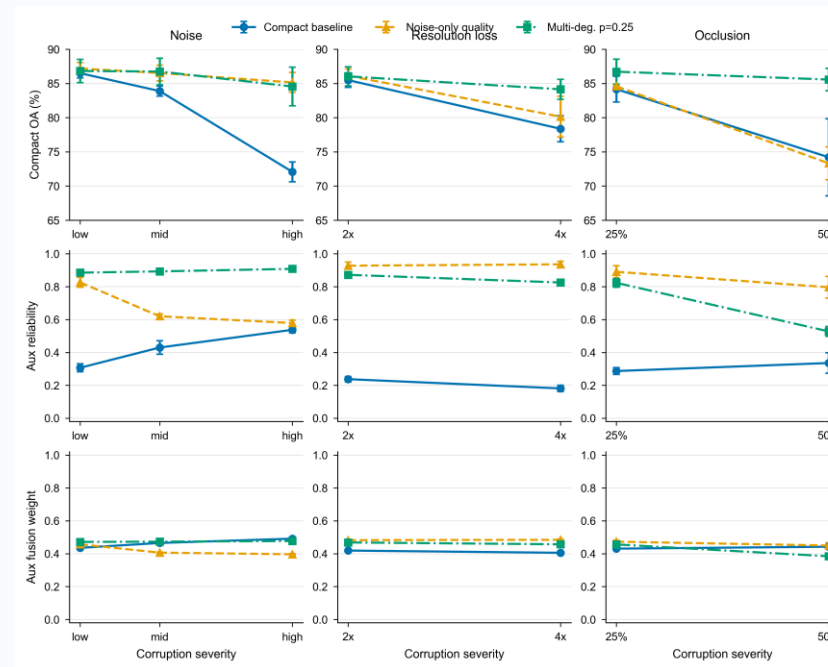
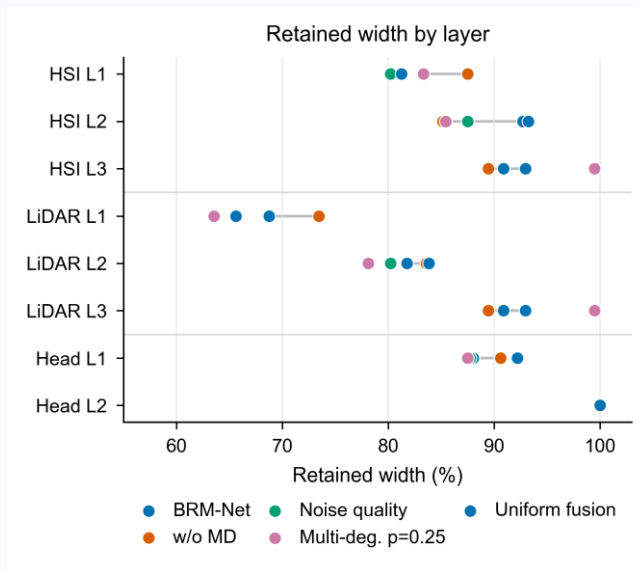
Conclusion

Robustness comes from degradation-supervised reliability, not from a reliability scalar alone.

Submission boundary

Routing, weighted CE, and profile label stabilization remain thesis extensions.

Visualization: retained width / reliability diagnostic / MUUFL error



Visuals should be read as mechanism diagnostics; reliability is degradation-aware, not physically calibrated.

Discussion: clean-robustness trade-off and limitations

Supported

Target MAC agreement and compact export are well supported on Houston2013.

Nuanced

MUUFLL improves robustness but sacrifices 1.64 pp clean full-modality OA.

Limited

Current datasets are still HSI-LiDAR style; HS-MS and HS-SAR remain future evidence.

A single full-modality OA is not enough for deployment-oriented evaluation.

Reliability weighting is useful only when trained with corrupted-but-available states.

The paper should avoid claims of real onboard deployment or physical sensor calibration.

Larger backbones and stronger lightweight baselines would strengthen submission impact.

Contribution summary

1

compact export

2

reliability fusion

3

joint evaluation

Fusion-compatible budgeted compact export

Learns gates and materializes compact models while preserving the fusion interface.

Degradation-supervised reliability fusion

Uses quality targets for corrupted-but-available modalities without reconstruction modules.

Accuracy-efficiency-robustness evaluation

Reports actual MACs, missing-modality states, degraded states, and multi-dataset evidence.

Future work

Extended thesis exploration

Dynamic routing, weighted CE, profile label stabilization, class-wise diagnostics.

Stronger evidence

Larger-backbone transfer, stronger lightweight baselines, denser corruption curves.

Demo system

Interactive dataset/state/budget selector for thesis defense demonstration.

Q&A

Main message: BRM-Net targets deployable compact multimodal classification under budget, missing-modality, and degraded-modality constraints.

TODO: final confusion matrix / final latency table / denser corruption grid remain outside this PPT's confirmed evidence.